

Ricky Ding

Software Engineer. Entrepreneur. Physics Student

Purpose of life: understand the human brain (NeuroAI), and read data from it (Connectomics)

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Education

National University of Singapore (NUS)

2016-2020

- BSc in *Physics*
- Grade: Honours, Highest Distinction
- Awards: Science & Technology Undergraduate Scholarship (4 years)
- Thesis: Density Potential Functional Theory in position and momentum space and its implementation in *PyTorch*
- [Download certificate](#) → [Verify certificate](#)

University of Göttingen, Germany

2019

- MSc courses in *Biophysics* (non-degree exchange program)
- Grade: 1.4 (German system: 1.0 best - 4.0 pass)
- Note: studied at *Max Planck Institute for Multidisciplinary Sciences*
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Skills

- **Languages:** Python; JavaScript (TypeScript); C++; Verilog; SQL
- **AI & Scientific:** *PyTorch*, Scikit-learn, Numba, NumPy
- **Frontend:** Vue3, React; Capacitor, React Native; ECharts
- **Backend:** Node.js, Bun, Elysia; FastAPI; PostgreSQL, pgvector, Redis, Caddy; Docker
- **Hardware:** MCU, FPGA/CPLD; Circuit & PCB Design; 3D Design

Employment

Center for Quantum Technologies @ NUS

Jul 2020 - May 2021

- Research Assistant in Software Development
- Worked on Superconducting Quantum Computing (Transmon):
 - Quantum Optimal Control using *Deep Reinforcement Learning (SAC)*
 - Controlling quantum hardware devices (microwave)
 - Electron Beam Lithography pattern generation (Python)
 - Temperature controller (PID)
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- Full-Stack Engineer, CTO, Co-founder
- Received investment from: Entrepreneur First (Pre-Seed, *USD 58K*), Cocoon Capital & Plug and Play (Seed, *USD 700K*)
- *Acquired* by a company in the USA
- [Featured in the BEAMP program](#) by Singapore's Building and Construction Authority and EnterpriseSG
- Worked on *AI for construction industry*:
 - Task duration prediction: *vector embedding, vector database* → *HDBSCAN clustering* → statistical computation
 - AI Planner: design project plans using *LLM*
 - Scheduling under complex constraints: Mixed Integer Linear Programming
 - Duration and Resource optimization: *Computational Graphs* → *Backpropagation* → *Gradient Descent (Adam)*

Open Source Projects

NeuroAI Research (*Neuroscience-Inspired AI*)

- [GitHub: Brain Modules](#)
 - Identify a list of brain's computational units/modules discovered in Neuroscience
 - Replicate these modules' functionalities using Artificial Neural Networks (ANN)
 - Purpose: to understand the brain and explore *new ANN architectures*
 - Keywords: *Place Cells, Transformer, GRU, MLP*
- [GitHub: Minimalist RL](#)
 - Minimalist and Decoupled *Reinforcement Learning algorithms*
 - Keywords: *SAC, PPO*
- [GitHub: Grad-Free Optim](#)
 - Gradient-free optimization of Neural Network parameters
 - Keywords: *Genetic Algorithm, Evolution Strategy*

Deep Learning for Trading

- [GitHub: Deep Trader](#)
 - Keywords: *Direct Policy Optimization, Differentiable Backtesting*
- [GitHub: Fast Trading Simulator](#)
 - *Numba* accelerated trading simulator
- [GitHub: Crypto Data Downloader](#)
 - High-speed crypto data downloader via concurrent API requests
- [GitHub: Crypto Monitor](#)
 - Crypto market monitor via WebSocket

Tesla Coil Research

- [GitHub: Tesla Coil Simulator](#)
 - Physics simulator for DRSSTC (Dual Resonant Solid State Tesla Coil), a coupled series RLC resonators system
- [GitHub: Musical Tesla Coil Gun](#)
 - Keywords: *Schematic & PCB Designs, Audio Synthesizer, CPLD, Phase-Shifted Full-Bridge*

Entrepreneurial Projects

EntreGames: *LLM* powered Social App

- Startup journey simulation with *AI generated stories*
- Social platform for entrepreneurs to connect
- Startup idea & founder screener for investors

Similar Minds: *LLM* powered Social App

- User share thoughts → *vector embeddings* → *similarity search* in *vector database* → people with similar thoughts

Evil AI Boss

- A *proactive anti-procrastination AI agent* that supervises your daily life
- It makes plans and assigns tasks to you, to ensure you stay on track in pursuing life's goal